

Software Requirements Specification

CocoConnect – A Digital Ecosystem for Coconut Estate Management

Version: 0.1

Status: Initial Draft

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Client: Coconut Growers Association of Sri Lanka

1. Introduction

1.1 Purpose

This document describes the initial software requirements identified for CocoConnect. CocoConnect is being developed as a digital platform to support coconut estate owners and other people involved in the coconut industry.

The main purpose of the system is to make some of the existing estate management and business activities easier to manage through one platform. At the moment, activities such as finding workers, finding buyers, selling coconut products and maintaining estate information can involve manual communication and separate methods.

CocoConnect is expected to bring these activities together in one system.

This is an initial draft of the requirements and some of the requirements may change after further discussions with the client and stakeholders.

1.2 Scope

CocoConnect will mainly focus on the coconut industry. The system will provide facilities for estate management, buying and selling products, finding workers and service providers, recording harvest information and communicating with other users.

The main users identified at this stage are Estate Owners, Estate Managers, Buyers, Sellers, Workers or Service Providers and Administrators.

The system will initially focus on coconut related activities but it should be possible to extend it to other agricultural products in the future.

1.3 Intended Users of this Document

This document will be used by the project team, project supervisors, client representatives, developers, designers and testers. It will also be useful when reviewing whether the system being developed matches the requirements identified during the initial analysis.

1.4 Terms Used

SRS means Software Requirements Specification.

Estate Owner refers to a person who owns and manages a coconut estate.

Estate Manager refers to a person who handles the daily activities of an estate.

Buyer refers to a person who is interested in purchasing products through the system.

Seller refers to a person who lists products for sale.

Worker or Service Provider refers to a person or business that provides services such as coconut plucking, husking, fertilizing or machinery related services.

Product Marketplace refers to the part of the system used for buying and selling products.

Workforce Marketplace refers to the part of the system used to find workers and service providers.

2. Overall Description

2.1 Product Perspective

CocoConnect is planned as a mobile and web based application. It will bring several activities related to coconut estate management into one platform.

The main areas identified for the system are user management, product marketplace, workforce marketplace, estate management, harvest management, ratings and reviews, notifications and administration.

The proposed technology stack includes Flutter for the frontend, Node.js and Express.js for the backend and PostgreSQL for data storage.

2.2 Main Functions

The initial functions identified for CocoConnect include:

User registration and login

Profile management

Estate registration

Estate information management

Product listing

Product searching

Finding workers and service providers

Sending and managing job requests

Recording harvest information

Viewing harvest history

Ratings and reviews

Notifications

Administrative management

2.3 User Roles

Estate Owner

The Estate Owner can manage estate information, view harvest information, find workers and service providers and list products for sale.

Estate Manager

The Estate Manager can manage the daily activities of an assigned estate and record relevant estate and harvest information.

Buyer

The Buyer can browse products, search for products and contact sellers.

Seller

The Seller can create and manage product listings and communicate with buyers.

Worker or Service Provider

Workers and service providers can create service profiles and receive requests from estate owners.

Administrator

The Administrator can manage users, monitor listings and handle administrative activities.

2.4 Operating Environment

The application is expected to support Android and iOS mobile devices as well as web browsers.

The proposed development environment includes Flutter, Node.js, Express.js and PostgreSQL.

2.5 Assumptions

Users are expected to have basic knowledge of using smartphones or computers.

Users are expected to provide correct information when registering and when creating product or service listings.

Most functions will require an internet connection.

2.6 Constraints

The system must protect user information.

Different users should only be able to access the functions relevant to their roles.

The application should be designed so that additional features can be added later.

3. Initial Functional Requirements

3.1 User Management

FR01: The system shall allow users to create an account.

FR02: The system shall allow users to log into their accounts.

FR03: The system shall allow users to log out.

FR04: The system shall allow users to update their profile information.

FR05: The system shall allow users to reset their password.

FR06: The system shall provide different access levels based on the user's role.

3.2 Product Marketplace

FR07: Sellers shall be able to create product listings.

FR08: Sellers shall be able to edit their listings.

FR09: Sellers shall be able to remove their listings.

FR10: Buyers shall be able to browse available products.

FR11: Buyers shall be able to search for products.

FR12: Buyers shall be able to view product details.

FR13: Buyers shall be able to contact sellers.

The initial product categories identified include DC coconut, curry coconut, coconut husk and other agricultural products that may be added later.

3.3 Workforce Marketplace

FR14: Workers and service providers shall be able to create service profiles.

FR15: Estate owners shall be able to search for workers and service providers.

FR16: Estate owners shall be able to send job requests.

FR17: Workers shall be able to accept or reject job requests.

FR18: The system shall maintain the status of job requests.

The initial service categories identified include coconut plucking, husking, fertilizing, coconut gathering, loading, tractor services and equipment supply.

3.4 Estate Management

FR19: Estate owners shall be able to register an estate.

FR20: Estate owners shall be able to update estate information.

FR21: Authorized users shall be able to view estate information.

FR22: The system shall store information related to registered estates.

3.5 Harvest Management

FR23: Authorized users shall be able to record harvest information.

FR24: The system shall store harvest records.

FR25: Authorized users shall be able to view previous harvest records.

3.6 Ratings and Reviews

FR26: Users shall be able to rate other users after completing a transaction or service.

FR27: Users shall be able to provide written reviews.

FR28: The system shall display user ratings.

3.7 Notifications

FR29: The system shall notify users about important activities.

FR30: Users shall receive notifications about job requests.

FR31: Users shall receive notifications when requests are accepted or rejected.

3.8 Administration

FR32: Administrators shall be able to manage user accounts.

FR33: Administrators shall be able to monitor product listings.

FR34: Administrators shall be able to review reported content.

FR35: Administrators shall be able to suspend users when necessary.

4. Non Functional Requirements

The system should provide a simple and easy to understand interface.

The system should respond within a reasonable amount of time during normal use.

User passwords and personal information should be protected.

The system should support multiple users.

The system should be designed so that additional agricultural products and features can be added in the future.

The system should be maintainable and the source code should be organized properly.

5. Business Rules

Only registered users can access protected features.

Users can only access functions allowed for their role.

Sellers should provide the required information when creating a product listing.

Workers should provide their service category when creating a service profile.

Job requests should be accepted before a service is considered assigned.

Ratings should only be provided after a completed transaction or service.

Estate Managers should only manage estates assigned to them.

Administrators can take action against accounts that violate platform rules.

6. Future Considerations

The following features were identified as possible future additions:

Online payments

GPS based services

Weather information

Additional agricultural products

Advanced analytics

Multilingual support

Push notifications

Equipment rental

Estate tourism services

AI based recommendations

7. Document Status

This document represents the initial requirements identified during the early stages of the CocoConnect project. The requirements will be reviewed with stakeholders and updated as the project progresses.